

DML Statements

Part 2

The MySQL LIMIT Clause

- The LIMIT clause is used to specify the number of records to return.
- The LIMIT clause is useful on large tables with thousands of records. Returning a large number of records can impact performance.

Syntax

- `SELECT` *column_name(s)*
`FROM` *table_name*
`WHERE` *condition*
`LIMIT` *number*;

The MySQL LIMIT Clause

```
1 • select * from actor
2 LIMIT 5;
```

```
4 • SELECT * FROM customer
5 WHERE active=1
6 LIMIT 5, 5;
```

00%  9:2

Result Grid   Filter Rows:

actor_id	first_name	last_name	last_update
1	PENELOPE	GUINNESS	2006-02-15 04:34:33
2	NICK	WAHLBERG	2006-02-15 04:34:33
3	ED	CHASE	2006-02-15 04:34:33
4	JENNIFER	DAVIS	2006-02-15 04:34:33
5	JOHNNY	LOLLOBRIGIDA	2006-02-15 04:34:33
NULL	NULL	NULL	NULL

[illegible]

The MySQL LIMIT Clause

```
14 -- What country has the 6th highest life expectancy?
15 * SELECT * FROM country
16 ORDER BY LifeExpectancy DESC
17 LIMIT 10;
18
```

Code	Name	Continent	Region	SurfaceArea	IndepYear	Population	LifeExpectancy	GDP	GDPOld	LocalName	GovernmentForm	Head
AND	Andorra	Europe	Southern Europe	468.00	1278	78000	83.5	1830.00	1830.00	Andorra	Parliamentary Coprincipality	Jan
MAC	Macao	Asia	Eastern Asia	18.00	1999	473000	81.6	5749.00	9940.00	Macao/Moken	Special Administrative Region of China	Jan
SMR	San Marino	Europe	Southern Europe	61.00	885	27000	81.1	510.00	1830.00	San Marino	Republic	1830
JPN	Japan	Asia	Eastern Asia	377825.00	-660	126714000	80.7	3787042.00	4192638.00	Nihon/Nippon	Constitutional Monarchy	Akih
SGP	Singapore	Asia	Southeast Asia	618.00	1965	3567000	80.1	86503.00	96318.00	Singapore/Singapura/Ninjapo/Singapur	Republic	Sell
AUS	Australia	Oceania	Australia and New Zealand	7741220.00	1901	18886000	79.8	351182.00	392911.00	Australia	Constitutional Monarchy, Federation	Elis
CHE	Switzerland	Europe	Western Europe	41284.00	1499	7160400	79.6	264478.00	256092.00	Schweiz/Suisse/Svizzera/Svizra	Federation	Adol
SWE	Sweden	Europe	Nordic Countries	449964.00	836	8861400	79.6	226452.00	227757.00	Sverige	Constitutional Monarchy	Carl
HKG	Hong Kong	Asia	Eastern Asia	1575.00	1997	6782000	79.5	166448.00	173610.00	Kiangong/Hong Kong	Special Administrative Region of China	Jon
CAN	Canada	North America	North America	9970610.00	1867	31147000	79.4	598862.00	625626.00	Canada	Constitutional Monarchy, Federation	Elis

Data Aggregations

What are data aggregations?

- In SQL, aggregate functions let you perform a calculation on multiple data and return a single value.
- That's why they are called “aggregate” functions.
- Those aggregate functions are `AVG()` , `COUNT()` , `SUM()` , `MIN()` , and `MAX()` .

Sample dataset

petstore

pet_id	pet_type	colour	name	cost
1	dog	black	Buster	899
2	cat	black	Morgan	699
3	cat	white	Diamond	1199
4	dog	golden	Dolly	1299
5	hamster	golden	Sabrina	49
6	hamster	brown	Sneak	99
7	dog	brown	Flo	799
8	fish	golden	Polly	19
9	dog	black	Lottie	599
10	cat	black	Felix	499

Data Aggregations

How many do
you have of
each pet?

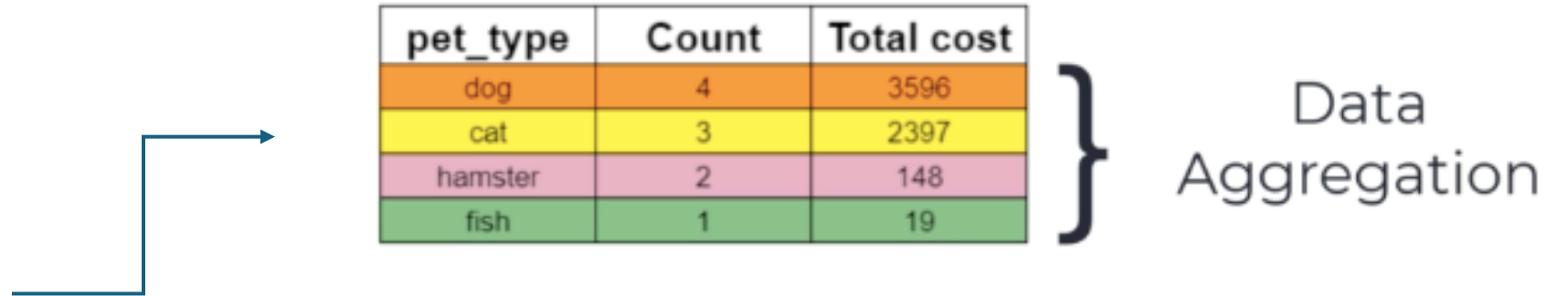
Sample questions that can
be answered using data
aggregations.

How much to
buy all of the
dogs?

pet_type	Count	Total cost
dog	4	3596
cat	3	2397
hamster	2	148
fish	1	19



Sample dataset



A blue line starts from the bottom left of the 'petstore' table, goes up, then right, ending in an arrow pointing to the 'pet_type' column of the summary table. To the right of the summary table is a large closing curly brace '}' followed by the text 'Data Aggregation'.

pet_type	Count	Total cost
dog	4	3596
cat	3	2397
hamster	2	148
fish	1	19

} Data Aggregation

petstore

pet_id	pet_type	colour	name	cost
1	dog	black	Buster	899
2	cat	black	Morgan	699
3	cat	white	Diamond	1199
4	dog	golden	Dolly	1299
5	hamster	golden	Sabrina	49
6	hamster	brown	Sneak	99
7	dog	brown	Flo	799
8	fish	golden	Polly	19
9	dog	black	Lottie	599
10	cat	black	Felix	499

Key Notes

1. Without performing an aggregation on your data, your query is pulling information back at the most granular level.
2. Data aggregations allow multiple values in your table to be grouped into one.
3. There are numerous aggregate functions we can use such as COUNT and SUM.

Key Points

- We use order by clause to get the summary of data in rows and is mostly taken in usage with the aggregate functions like Count, Sum, etc.
- An aggregate function includes values grouped together to form a single value that provides a summary of the data list.

Group By Clause

- The GROUP BY statement groups rows that have the same values into summary rows, like "find the number of customers in each country".
- The GROUP BY statement is often used with aggregate functions (COUNT(), MAX(), MIN(), SUM(), AVG()) to group the result-set by one or more columns.
- We use Group by clause to get the groups present in data.

GROUP BY Syntax

- `SELECT column_name(s)`
`FROM table_name`
`WHERE condition`
`GROUP BY column_name(s)`
`ORDER BY column_name(s);`

GROUP BY Syntax

- `SELECT column_name(s)`
`FROM table_name`
`WHERE condition`
`GROUP BY column_name(s)`
`ORDER BY column_name(s);`
- `SELECT COUNT(age), age`
- `FROM employee`
- `GROUP BY age ;`

Group By Clause

- The GROUP BY statement groups rows that have the same values into summary rows, like "find the number of customers in each country".
- The GROUP BY statement is often used with aggregate functions (COUNT(), MAX(), MIN(), SUM(), AVG()) to group the result-set by one or more columns.

Examples:

```
180 • SELECT pet_type
181     FROM petStore
182     GROUP BY pet_type;
```

100%	↕	23:177	
Result Grid   Filter Rows: 			
	pet_type		
▶	dog		
▬	cat		
	hamster		
▬	fish		

Select Distinct

- The SELECT DISTINCT statement is used to return only distinct (different) values.

```
180 • SELECT pet_type
181 FROM petStore
182 GROUP BY pet_type;
```

100% 1:176

Result Grid Filter Rows:

pet_type
dog
cat
hamster
fish



```
184 • SELECT DISTINCT pet_type
185 FROM petStore;
```

100% 11:180

Result Grid Filter Rows: Search

pet_type
dog
cat
hamster
fish

Group By Clause (SUM)

How much is the total sales if all pets are sold?

```
187 • SELECT * FROM petStore;
```

100%

7:182

Result Grid

Filter Rows:

Search

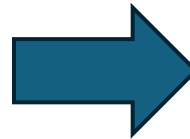
	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499

Group By Clause (SUM)

- How much is the total sales per pet type?

```
189 • SELECT SUM(cost)
190 FROM petStore
191 GROUP BY pet_type;
192
```

100%	↕	1:189
Result Grid   Filter Rows:		
	SU...	
▶	3596	
	2397	
	148	
	19	



```
189 • SELECT pet_type, SUM(cost)
190 FROM petStore
191 GROUP BY pet_type;
```

100%	↕	12:187
Result Grid   Filter Rows: Search		
	pet_type	SU...
▶	dog	3596
	cat	2397
	hamster	148
	fish	19

Group By Clause (Counting Records)

1. Counting all rows of data

Example 1

```
SELECT COUNT(*)  
FROM customer;
```

Example 2

```
SELECT Gender,  
       COUNT(*)  
FROM customer  
GROUP BY Gender;
```

Group By Clause (Counting Records)

2. Counting non-null values in a column

Example 1

```
SELECT COUNT(Age)  
FROM customer;
```

Example 2

```
SELECT Region,  
       COUNT(LifeExp)  
FROM country  
GROUP BY Region;
```

Group By Clause (Counting Records)

3. Counting distinct values in a column

Example 1

```
SELECT COUNT(DISTINCT Town)  
FROM customer;
```

Example 2

```
SELECT Gender,  
       COUNT(DISTINCT Model)  
FROM cars  
GROUP BY Gender;
```

Group By Clause (Counting Records)

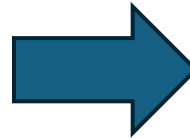
- Count the records in the dataset.

187 • `SELECT * FROM petStore;`

100%  1:183

Result Grid   Filter Rows:

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499



193 • `SELECT COUNT(*) FROM petStore;`

100%	1:188
Result Grid  Filter Rows: Search	
	C...
▶	10

Group By Clause (Counting Records)

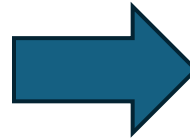
- Count the records in the dataset.

205 • `SELECT * FROM petStore;`

100% 1:199

Result Grid Filter Rows: Search

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499
	11	dog	black		299



200 • `SELECT pet_type,`
201 `COUNT(*)`
202 `FROM petStore`
203 `GROUP BY pet_type;`

100% 13:197

Result Grid Filter Rows: Search

	pet_type	COUNT(*)
▶	dog	5
	cat	3
	hamster	2
	fish	1

Group By Clause (Counting Records)

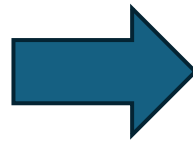
- Count the records in the dataset.
- Use COUNT(*) when you count all records in the table.
- Use COUNT(column_name) if you want to count data in the specific column.

193 • `SELECT COUNT(*) FROM petStore;`

100% 13:190

Result Grid Filter Rows: Search

C...
10



195 • `SELECT COUNT(pet_type) FROM petStore;`

100% 11:195

Result Grid Filter Rows: Search Ex

COUNT(pet_typ...
10

Group By Clause (Counting Records)

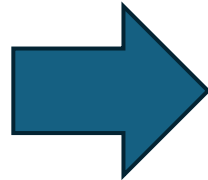
- How many cats are in the record?

```
187 • SELECT * FROM petStore;
```

100% 1:183

Result Grid   Filter Rows:

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499



```
197 • SELECT COUNT(pet_type) FROM petStore
```

```
198 WHERE pet_type="cat";
```

```
199
```

100% 1:188

Result Grid





Filter Rows:

 Search

E

COUNT(pet_typ...
3

Group By Clause (Counting Records)

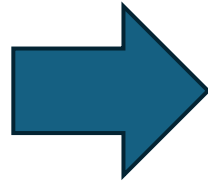
- How many records of pets are there?

```
207 • SELECT COUNT(pet_type)
208     FROM petStore;
209
```

100% 15:208

Result Grid Filter Rows: Search

COUNT(pet_typ...
11



```
207 • SELECT pet_type, COUNT(pet_type)
208     FROM petStore
209     GROUP BY pet_type;
```

100% 1:204

Result Grid Filter Rows: Search

pet_type	COUNT(pet_typ...
dog	5
cat	3
hamster	2
fish	1

Group By Clause (Counting Records)

- How many unique pets are there?

```
218 • SELECT COUNT(DISTINCT pet_type)
219 FROM petStore;
```

100%	↕	3:214
Result Grid   Filter Rows: Search		
COUNT(DISTINCT pet_typ...		
▶	4	

Group By Clause (SUM)

- The SUM() function returns the total sum of a numeric column.
- `SELECT SUM(column_name)
FROM table_name
WHERE condition;`

Group By Clause (SUM)

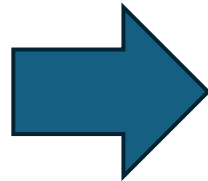
- What is the total sales if all cats are sold?

225 • **SELECT** ***FROM** petStore;

100% 19:223

Result Grid Filter Rows: Search

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499
	11	dog	black		299



227 • **SELECT** SUM(cost) **as** 'CAT SALES'

228 **FROM** petStore

229 **WHERE** pet_type="cat";

230

100% 25:227

Result Grid Filter Rows: Search

	CAT SALES
	2397

Group By Clause (SUM)

- Add a WHERE Clause
- You can add a WHERE clause to specify conditions.
- Example: Return the sum of the Quantity field for the product with ProductID 11:

- Sample Code:

```
SELECT SUM(Quantity)
FROM OrderDetails
WHERE ProductId = 11;
```


Group By Clause (SUM)

- Use an Alias
- Give the summarized column a name by using the AS keyword.

Example:

Name the column "total":

Sample Code:

- `SELECT SUM(Quantity) AS total
FROM OrderDetails;`

Group By Clause (SUM)

- Use SUM() with GROUP BY
- Here we use the SUM() function and the GROUP BY clause, to return the Quantity for each OrderID in the OrderDetails table:

Example:

- `SELECT OrderID, SUM(Quantity) AS [Total Quantity]
FROM OrderDetails
GROUP BY OrderID;`

Group By Clause (SUM)

- SUM() With an Expression
- The parameter inside the SUM() function can also be an expression.
- If we assume that each product in the OrderDetails column costs 10 dollars, we can find the total earnings in dollars by multiply each quantity with 10:
- `SELECT SUM(Quantity * 10)
FROM OrderDetails;`

Group By Clause (SUM)

- Example:

```
---
227 • SELECT SUM(cost*2) as 'CAT SALES'
228 FROM petStore
229 WHERE pet_type="cat";
230
```

100%	↺	23:225	
Result Grid   Filter Rows: Search			
	CAT SALES		
▶	4794		

The SQL MIN() and MAX() Functions

- The MIN() function returns the smallest value of the selected column.
- The MAX() function returns the largest value of the selected column.
- Find the lowest price in the Price column:

The SQL MIN() and MAX() Functions

- `SELECT MIN(column_name)
FROM table_name
WHERE condition;`
- `SELECT MAX(column_name)
FROM table_name
WHERE condition;`

The SQL MIN() and MAX() Functions

- Find the lowest price in the Price column:
- `SELECT MIN(Price)`
`FROM Products;`
- Find the highest price in the Price column:
- `SELECT MAX(Price)`
`FROM Products;`

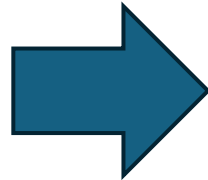
The SQL MAX() Function

245 • **SELECT** * **FROM** petStore;

100% 10:242

Result Grid Filter Rows: Search

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499
	11	dog	black		299



247 • **SELECT** max(cost)

248 **FROM** petStore;

100% 13:242

Result Grid Filter Rows:

	max(cost)
▶	1299

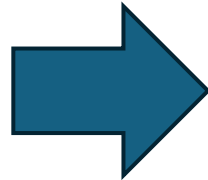
The SQL MIN() Function

245 • **SELECT** * **FROM** petStore;

100% 10:242

Result Grid Filter Rows: Search

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499
	11	dog	black		299



250 • **SELECT** MIN(cost)

251 **FROM** petStore;

100% 11:245

Result Grid Filter Rows:

	MIN(cost)
▶	19

The SQL MIN() and MAX() Functions

- Note: When you perform MIN and MAX function, ensure that the data type of the column is of a numeric type (ex. int).
- Otherwise, MIN and MAX function will generate incorrect result.

Examples

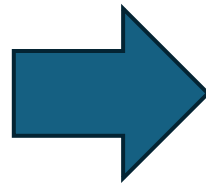
- What is the price of the most expensive dog?

245 • `SELECT * FROM petStore;`

100% 10:242

Result Grid Filter Rows: Search

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499
	11	dog	black		299



253 • `SELECT max(cost)`
254 `FROM petStore`
255 `WHERE pet_type="dog";`

100% 1:249

Result Grid Filter Rows: Search

	max(cost)
▶	1299

Examples

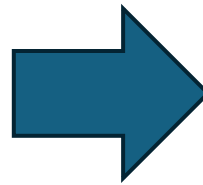
- What is the price of the least expensive cat?

245 • `SELECT * FROM petStore;`

100% 10:242

Result Grid Filter Rows: Search

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499
	11	dog	black		299



257 • `SELECT MIN(cost)`
258 `FROM petStore`
259 `WHERE pet_type="CAT";`

100% 13:253

Result Grid Filter Rows: Search

	MIN(cost)
▶	499

Examples

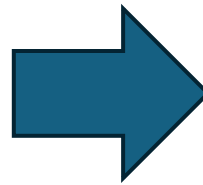
- What is the price of the most expensive and least expensive dog?

245 • `SELECT * FROM petStore;`

100% 10:242

Result Grid Filter Rows: Search

	pet_id	pet_type	colour	name	cost
▶	1	dog	black	Buster	899
	2	cat	black	Morgan	699
	3	cat	white	Diamond	1199
	4	dog	golden	Dolly	1299
	5	hamster	golden	Sabrina	49
	6	hamster	brown	Sneak	99
	7	dog	brown	Flo	799
	8	fish	golden	Polly	19
	9	dog	black	Lotte	599
	10	cat	black	Felix	499
	11	dog	black		299



261 • `SELECT MIN(cost), MAX(cost)`
262 `FROM petStore`
263 `WHERE pet_type="CAT";`

100% 18:257

Result Grid Filter Rows: Search

	MIN(cost)	MAX(cost)
▶	499	1199

The SQL AVG() Function

- The AVG() function returns the average value of a numeric column.

Syntax:

- ```
SELECT AVG(column_name)
FROM table_name
WHERE condition;
```

# The SQL AVG() Function

- The AVG() function returns the average value of a numeric column.

Example:

- Find the average price of all products:
- `SELECT AVG(Price)`  
`FROM Products;`

# Example:

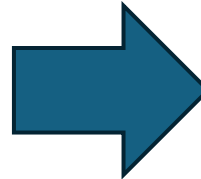
What is the average cost of the pets?

```
245 • SELECT * FROM petStore;
```

100%  10:242

Result Grid   Filter Rows:

|   | pet_id | pet_type | colour | name    | cost |
|---|--------|----------|--------|---------|------|
| ▶ | 1      | dog      | black  | Buster  | 899  |
| ▶ | 2      | cat      | black  | Morgan  | 699  |
| ▶ | 3      | cat      | white  | Diamond | 1199 |
| ▶ | 4      | dog      | golden | Dolly   | 1299 |
| ▶ | 5      | hamster  | golden | Sabrina | 49   |
| ▶ | 6      | hamster  | brown  | Sneak   | 99   |
| ▶ | 7      | dog      | brown  | Flo     | 799  |
| ▶ | 8      | fish     | golden | Polly   | 19   |
| ▶ | 9      | dog      | black  | Lotte   | 599  |
| ▶ | 10     | cat      | black  | Felix   | 499  |
| ▶ | 11     | dog      | black  |         | 299  |



```
233 • SELECT AVG(cost)
234 FROM petStore;
```

|                                                                                                                                                                                                         |           |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------|
| 100%                                                                                                                                                                                                    | ↕         | 14:228 |
| <b>Result Grid</b>   Filter Rows: |           |        |
|                                                                                                                                                                                                         | AVG(cost) |        |
| ▶                                                                                                                                                                                                       | 587.1818  |        |



# Example:

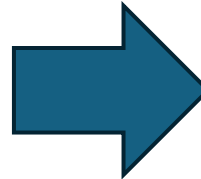
- What is the average cost of cats?

```
245 • SELECT * FROM petStore;
```

100%  10:242

Result Grid   Filter Rows:

|   | pet_id | pet_type | colour | name    | cost |
|---|--------|----------|--------|---------|------|
| ▶ | 1      | dog      | black  | Buster  | 899  |
|   | 2      | cat      | black  | Morgan  | 699  |
|   | 3      | cat      | white  | Diamond | 1199 |
|   | 4      | dog      | golden | Dolly   | 1299 |
|   | 5      | hamster  | golden | Sabrina | 49   |
|   | 6      | hamster  | brown  | Sneak   | 99   |
|   | 7      | dog      | brown  | Flo     | 799  |
|   | 8      | fish     | golden | Polly   | 19   |
|   | 9      | dog      | black  | Lotte   | 599  |
|   | 10     | cat      | black  | Felix   | 499  |
|   | 11     | dog      | black  |         | 299  |



```
265 • SELECT AVG(cost)
266 FROM petStore
267 WHERE pet_type="CAT";
```

100%  14:262

Result Grid   Filter Rows:

|   | AVG(cost) |  |
|---|-----------|--|
| ▶ | 799.0000  |  |

# Example:

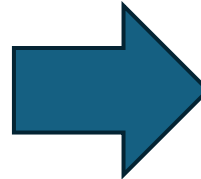
- What is the average cost of hamsters, cats, fish, and dogs?

245 • `SELECT * FROM petStore;`

100% 10:242

Result Grid Filter Rows: Search

|   | pet_id | pet_type | colour | name    | cost |
|---|--------|----------|--------|---------|------|
| ▶ | 1      | dog      | black  | Buster  | 899  |
|   | 2      | cat      | black  | Morgan  | 699  |
|   | 3      | cat      | white  | Diamond | 1199 |
|   | 4      | dog      | golden | Dolly   | 1299 |
|   | 5      | hamster  | golden | Sabrina | 49   |
|   | 6      | hamster  | brown  | Sneak   | 99   |
|   | 7      | dog      | brown  | Flo     | 799  |
|   | 8      | fish     | golden | Polly   | 19   |
|   | 9      | dog      | black  | Lotte   | 599  |
|   | 10     | cat      | black  | Felix   | 499  |
|   | 11     | dog      | black  |         | 299  |



265 • `SELECT pet_type, AVG(cost)`  
266 `FROM petStore`  
267 `GROUP BY pet_type;`

100% 22:263

Result Grid Filter Rows: Search

|   | pet_type | AVG(cost) |
|---|----------|-----------|
| ▶ | dog      | 779.0000  |
|   | cat      | 799.0000  |
|   | hamster  | 74.0000   |
|   | fish     | 19.0000   |

# Example:

- What is the average cost of hamsters, cats, fish, and dogs, from the most expensive average to the least expensive average?

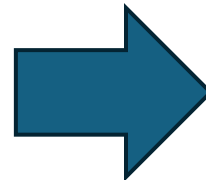
```
245 • SELECT * FROM petStore;
```

100%10:242

Result GridFilter Rows:

Search

|   | pet_id | pet_type | colour | name    | cost |
|---|--------|----------|--------|---------|------|
| ▶ | 1      | dog      | black  | Buster  | 899  |
|   | 2      | cat      | black  | Morgan  | 699  |
|   | 3      | cat      | white  | Diamond | 1199 |
|   | 4      | dog      | golden | Dolly   | 1299 |
|   | 5      | hamster  | golden | Sabrina | 49   |
|   | 6      | hamster  | brown  | Sneak   | 99   |
|   | 7      | dog      | brown  | Flo     | 799  |
|   | 8      | fish     | golden | Polly   | 19   |
|   | 9      | dog      | black  | Lotte   | 599  |
|   | 10     | cat      | black  | Felix   | 499  |
|   | 11     | dog      | black  |         | 299  |



```
265 • SELECT pet_type, AVG(cost)
266 FROM petStore
267 GROUP BY pet_type
268 ORDER BY AVG(cost) DESC;
```

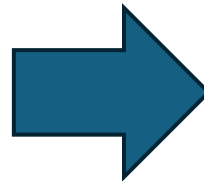
| 100%                                                                                                                                                                                                                                 | ↕        | 29:261    |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|--|
| Result Grid   Filter Rows: <input type="text" value="Search"/> |          |           |  |
|                                                                                                                                                                                                                                      | pet_type | AVG(cost) |  |
| ▶                                                                                                                                                                                                                                    | cat      | 799.0000  |  |
|                                                                                                                                                                                                                                      | dog      | 779.0000  |  |
|                                                                                                                                                                                                                                      | hamster  | 74.0000   |  |
|                                                                                                                                                                                                                                      | fish     | 19.0000   |  |

# Example:

- What's the difference?
- In the left example, the average is sorted according to pet type in descending order.
- While the data on the right side, the average is sorted according to the average cost in descending order.

```
265 • SELECT pet_type, AVG(cost)
266 FROM petStore
267 GROUP BY pet_type
268 ORDER BY pet_type DESC;
269
```

| 100%                            | 22:263    |
|---------------------------------|-----------|
| Result Grid Filter Rows: Search |           |
| pet_type                        | AVG(cost) |
| hamster                         | 74.0000   |
| fish                            | 19.0000   |
| dog                             | 779.0000  |
| cat                             | 799.0000  |



```
265 • SELECT pet_type, AVG(cost)
266 FROM petStore
267 GROUP BY pet_type
268 ORDER BY AVG(cost) DESC;
```

| 100%                            | 29:261    |
|---------------------------------|-----------|
| Result Grid Filter Rows: Search |           |
| pet_type                        | AVG(cost) |
| cat                             | 799.0000  |
| dog                             | 779.0000  |
| hamster                         | 74.0000   |
| fish                            | 19.0000   |

# Key Notes on GROUP BY

1. The GROUP BY clause is used to group rows that have the same values
2. The GROUP BY clause is included in SELECT statements.
3. The GROUP BY clause is included in the SELECT statement when aggregate calculations are performed on your data to generate summary tables.

# Activity: (DML Statements Part 1)

1. Sort the pets record according to pet type.
2. Display pet names that starts with D.
3. Display pet names that starts with Sa.
4. Display all records for for fish and hamster only.
5. Sort all records from the most to least expensive pets.

|

# Activity: (DML Statements Part 1)

- Answers:

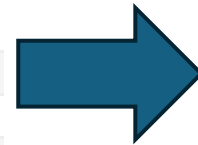
1. Sort the pet records according to pet type.

```
163 • SELECT * FROM petStore;
```

100% 3:161

Result Grid Filter Rows: Search

|   | pet_id | pet_type | colour | name    | cost |
|---|--------|----------|--------|---------|------|
| ▶ | 1      | dog      | black  | Buster  | 899  |
|   | 2      | cat      | black  | Morgan  | 699  |
|   | 3      | cat      | white  | Diamond | 1199 |
|   | 4      | dog      | golden | Dolly   | 1299 |
|   | 5      | hamster  | golden | Sabrina | 49   |
|   | 6      | hamster  | brown  | Sneak   | 99   |
|   | 7      | dog      | brown  | Flo     | 799  |
|   | 8      | fish     | golden | Polly   | 19   |
|   | 9      | dog      | black  | Lotte   | 599  |
|   | 10     | cat      | black  | Felix   | 499  |



```
165 • SELECT * FROM petStore
```

```
166 ORDER BY pet_type;
```

100% 1:162

Result Grid Filter Rows: Search

|   | pet_id | pet_type | colour | name    | cost |
|---|--------|----------|--------|---------|------|
|   | 2      | cat      | black  | Morgan  | 699  |
|   | 3      | cat      | white  | Diamond | 1199 |
|   | 10     | cat      | black  | Felix   | 499  |
|   | 1      | dog      | black  | Buster  | 899  |
|   | 4      | dog      | golden | Dolly   | 1299 |
|   | 7      | dog      | brown  | Flo     | 799  |
|   | 9      | dog      | black  | Lotte   | 599  |
|   | 8      | fish     | golden | Polly   | 19   |
|   | 5      | hamster  | golden | Sabrina | 49   |
| ▶ | 6      | hamster  | brown  | Sneak   | 99   |

# Activity: (DML Statements Part 1)

- Answers:

2. Display the pet names that starts with D.

```
168 • SELECT * FROM petStore
169 WHERE name LIKE 'D%';
```

100%

22:165

Result Grid

Filter Rows:

Search

|  | pet_id | pet_type | colour | name    | cost |
|--|--------|----------|--------|---------|------|
|  | 3      | cat      | white  | Diamond | 1199 |
|  | 4      | dog      | golden | Dolly   | 1299 |



# Activity: (DML Statements Part 1)

- Answers:

3. Display the pet names that start with Sa.

```
171 • SELECT * FROM petStore
172 WHERE name LIKE 'Sa%';
```

100% 18:169

**Result Grid**   Filter Rows:

| pet_id | pet_type | colour | name    | cost |
|--------|----------|--------|---------|------|
| 5      | hamster  | golden | Sabrina | 49   |

# Activity: (DML Statements Part 1)

- Answers:

4. Display all records for fish and hamster only.

```
174 • SELECT * FROM petStore
175 WHERE NOT pet_type= "dog" AND NOT pet_type= "cat";
```

100%

1:170

Result Grid

Filter Rows:

Search

Export:

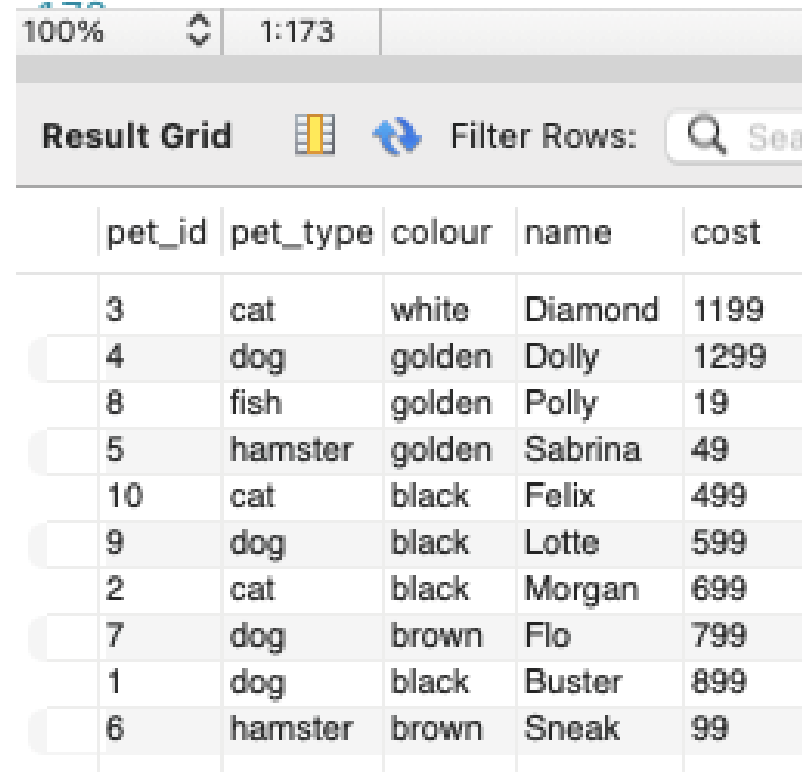
|  | pet_id | pet_type | colour | name    | cost |  |
|--|--------|----------|--------|---------|------|--|
|  | 5      | hamster  | golden | Sabrina | 49   |  |
|  | 6      | hamster  | brown  | Sneak   | 99   |  |
|  | 8      | fish     | golden | Polly   | 19   |  |
|  |        |          |        |         |      |  |

# Activity: (DML Statements Part 1)

- Answers:

5. Sort all records from the most to least expensive pets.

```
177 • SELECT * FROM petStore
178 ORDER BY cost;
```



100% 1:173

Result Grid Filter Rows: Search

|  | pet_id | pet_type | colour | name    | cost |
|--|--------|----------|--------|---------|------|
|  | 3      | cat      | white  | Diamond | 1199 |
|  | 4      | dog      | golden | Dolly   | 1299 |
|  | 8      | fish     | golden | Polly   | 19   |
|  | 5      | hamster  | golden | Sabrina | 49   |
|  | 10     | cat      | black  | Felix   | 499  |
|  | 9      | dog      | black  | Lotte   | 599  |
|  | 2      | cat      | black  | Morgan  | 699  |
|  | 7      | dog      | brown  | Flo     | 799  |
|  | 1      | dog      | black  | Buster  | 899  |
|  | 6      | hamster  | brown  | Sneak   | 99   |

# Activity: Refer to the SALES table

| sale_id | product_name | category    | quantity | unit_price | total_price |
|---------|--------------|-------------|----------|------------|-------------|
| 1       | Laptop       | Electronics | 3        | 800        | 2400        |
| 2       | Headphones   | Electronics | 5        | 100        | 500         |
| 3       | Coffee Maker | Home Goods  | 2        | 50         | 100         |
| 4       | Phone        | Electronics | 4        | 500        | 2000        |
| 5       | Blender      | Home Goods  | 6        | 40         | 240         |
| 6       | Laptop       | Electronics | 2        | 800        | 1600        |
| 7       | Headphones   | Electronics | 8        | 100        | 800         |
| 8       | Coffee Maker | Home Goods  | 1        | 50         | 50          |
| 9       | Phone        | Electronics | 2        | 500        | 1000        |
| 10      | Blender      | Home Goods  | 3        | 40         | 120         |

1. What is the most expensive laptop?
2. How many products are there?
3. How many coffee makers are there?
4. How many categories are there?
5. Display products under electronics category.
6. What is the average quantity.